Inverted Shells: The Spheres of Dharma Turned Inside Out



Polar reciprocation reverses the Spheres of Dharma into a new exact φ³ cycle of nested icosahedral shells: one kind of piece per layer, an evenly stepped nest, the great stellated dodecahedron locked in the outer gap, a dual pair in the inner gap, the golden frame, the cycle repeating inward to its core, and the checks that keep each step exact

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Draft 4 · 16 October 2026 · DICTO DOI: on publication · Licence CC BY 4.0 · Data, scripts, exact checks and renders are included with this record and at https://github.com/DICTOR-Master/kaleidohedra (branch research: studies/2026-10-16-polar-dharma and studies/2026-10-16-flip-test; shared checks in studies/_lib).
The Spheres of Dharma, the Hecatodihedron, the idea of turning the cycle inside out by polar reciprocation, the call for an even nest, the stars and the Hecatodihedron tried in the gaps, the dual pair and its reversed order, the golden rectangles, the endless core, the flip test, the demand for stricter checks and the name Ogdoecontahedron are DICTO's. Computation, exact arithmetic and renders were done with Claude (Anthropic) as a research assistant.

Abstract. The Spheres of Dharma nest six icosahedral solids exactly, from an icosahedron out to the icosahedron φ³ larger; between the truncated icosahedron and the triacontahedron, the layer closes part-way into the DICTO Hecatodihedron. DICTO asked what the cycle looks like turned inside out. Turning each layer's pieces over does not work: no layer closes into a shell. Taking the six solids and the Hecatodihedron stage as seven stations and replacing each by its polar reciprocal reverses the nesting and gives a new exact cycle: dodecahedron ⊂ icosidodecahedron ⊂ **Ogdoecontahedron** ⊂ pentakis dodecahedron ⊂ icosahedron ⊂ rhombic triacontahedron ⊂ dodecahedron φ³ larger (volume ratio φ⁹). Its one new station, the dual of the Hecatodihedron, is the DICTO Ogdoecontahedron (DICTO DOI: 10.5281/zenodo.23404688). Cutting each layer by cones over the faces of its inner solid gives **one kind of piece per layer** (12, 12, 20, 60, 20, 30), every volume exact in ℚ(√5) with remainder 0. The cycle inherits two loose ends. Tightening them gives an evenly stepped nest, still one kind per layer and exact. Left loose, the large outer gap holds the great stellated dodecahedron at exactly one size: its 20 points on the outer dodecahedron's corners, its 12 valleys on the triacontahedron's tips, cutting the gap into 60 tetrahedra and 60 pentagonal pyramids. The inner gap admits no star and no locked dual pair among those tested, but the truncated icosidodecahedron pair fits in reverse order (as polarity demands), with play exactly √5·φ³/9. The golden rectangles that frame the whole are the outer dodecahedron's own small stellation, and the cycle repeats inward without end. We also record the checks this study forced on us, because three of our first tables were wrong.

1. The Spheres of Dharma turned inside out

The Spheres of Dharma (DICTO DOI: 10.5281/zenodo.23310833) is an exact, endlessly repeating nesting of six icosahedral solids, the Ritsudō cycle: icosahedron ⊂ icosidodecahedron ⊂ dodecahedron ⊂ truncated icosahedron ⊂ rhombic triacontahedron ⊂ icosahedron φ³ larger, each layer filled by a few repeated pieces. In its truncated icosahedron → triacontahedron layer the hexagonal caps close part-way into a convex solid, the **DICTO Truncated-Icosahedral Hecatodihedron** (DICTO DOI: 10.5281/zenodo.23360858): the

truncated icosahedron with a Shield on each hexagon, 102 faces (12 pentagons, 30 rhombi, 60 triangles) and 80 corners. It is the seventh stage of the cycle when each closed stage is taken as one solid (Capturing the In-betweens, DICTO DOI: 10.5281/zenodo.23345502). Here we take the six solids and the Hecatodihedron stage as seven stations.

First attempt: flip the pieces. DICTO's question (13 October 2026) was what happens to the cycle turned inside out. The direct way is to turn every piece of every layer (skin) over, as a physical half-turn about a line through the point on the piece's own axis midway between its two solids; every Dharma piece has a mirror plane through its axis, so this is the same as reflecting it in the midway plane. No skin closes into a shell (Table 1, float). Skins of a few large pyramids drift apart, their points now turned inward; the closed skins crowd, about a third of the midway surface covered twice. In every skin the flipped pieces rest on the inner solid and never cut into it. Dharma's pieces are shaped for their direction, so turning the cycle inside out needs new pieces.

Skin	Pieces	Midway surface covered	Covered twice	Deepest overlap
icosahedron → icosidodecahedron	32	100%	35.1%	0.369
icosidodecahedron → dodecahedron	20	3.1%	0	none (pieces apart)
dodecahedron → truncated icosahedron	42	67.1%	4.4%	0.163
truncated icosahedron → Hecatodihedron	20	7.7%	0	none
Hecatodihedron → triacontahedron	72	17.8%	0.1%	0.070
triacontahedron → icosahedron φ^3	50	100%	36.4%	0.309

Table 1. The flip test (float; cover of the midway surface, halfway between the two solids along each direction, sampled at 4000 directions; deepest overlap between neighbouring flipped pieces, in the units of the Dharma coordinates).

Second attempt: polar reciprocation. Polar reciprocation in a sphere about the common centre sends every plane to a point and every point to a plane; a solid goes to its dual, and inclusion reverses: if $P \subset Q$ then $Q^* \subset P^*$ (a classical fact of convex geometry; see e.g. Firsching 2015). So the reversed cycle is again a nest, read from the other end:

$$\mathbf{D \subset ID \subset Ogdoecontahedron \subset PD \subset I \subset RT \subset D \cdot \varphi^3}$$

(dodecahedron, icosidodecahedron, the Hecatodihedron's dual, pentakis dodecahedron, icosahedron, rhombic triacontahedron, dodecahedron). We computed every pole exactly in $\mathbb{Q}(\sqrt{5})$ from the snapped original corners. The outer dodecahedron is exactly φ^3 times the inner one, so the volume closes at $38 + 17\sqrt{5} = \varphi^9$. Duality swaps solids in pairs, so the icosidodecahedron and the triacontahedron, and the icosahedron and the dodecahedron, appear in both cycles in swapped places; the truncated icosahedron becomes the pentakis dodecahedron, and the Hecatodihedron becomes something new.

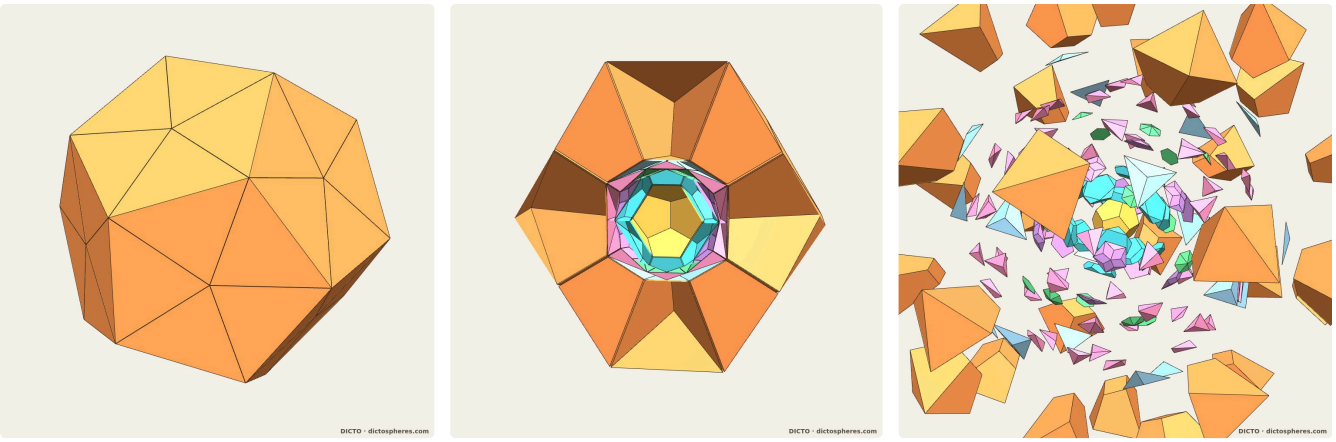


Figure 1. The reversed cycle, faithful to the original: (a) closed, the outer dodecahedron; (b) cut in half: under reciprocation the large layers become small and the small large, so the outer layer is far the largest; (c) pulled apart by layer.

2. The new station

The polar of the Hecatodihedron is a new solid, which DICTO named the **Ogdoecontahedron** for its 80 faces: 20 regular hexagons and 60 kites, with 102 corners and 180 edges. The solid itself, including how it sits between the icosidodecahedron and the pentakis dodecahedron, is described in *The DICTO Ogdoecontahedron* (DICTO DOI: 10.5281/zenodo.23404688).

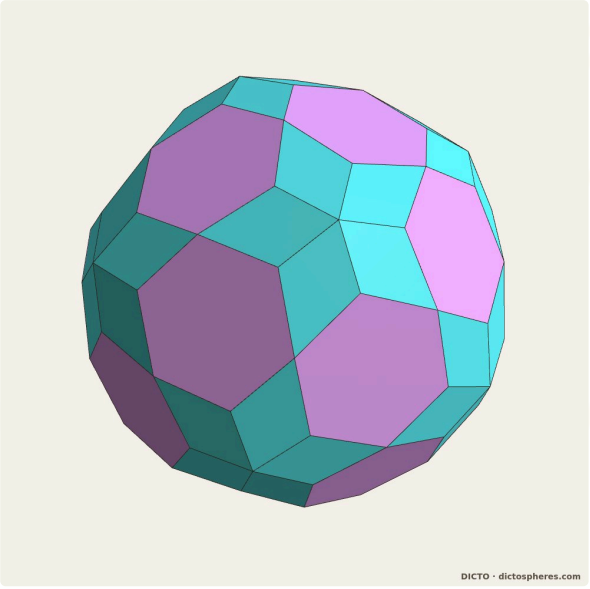


Figure 2. The DICTO Ogdoecontahedron, the new station: hexagons wisteria, kites cyan.

3. One kind of piece per layer

We cut each layer A ⊂ B by the cones from the centre over the faces of the inner solid A: each piece is cone(F) ∩ B beyond F, convex by construction (the Kion construction of Theaetetus, DICTO DOI: 10.5281/zenodo.23378070). Faces of A that already lie on B's surface give no piece. Every corner was solved exactly as the meeting point of three exact planes, and every volume summed exactly against its layer.

Layer	Pieces	Each piece	Exact volume
D → ID	12	a column on each dodecahedron face (two regular pentagons, 5 pentagonal and 5 triangular sides)	(−144185 + 64487√5)/243
ID → Ogdoecontahedron	12	a cap on each icosidodecahedron pentagon (the icosidodecahedron's triangles lie in the hexagons and give no piece)	(−155 + 101√5)/4617
Ogdoecontahedron → PD	20	a low pyramid on each regular hexagon (the kites lie in PD's faces)	(−22 + 10√5)/81
PD → I	60	an irregular wedge, 5 corners and 6 triangles	(−6560 + 2936√5)/513
I → RT	20	a triangular pyramid on each icosahedron face	(−796 + 356√5)/3
RT → D·φ³	30	a roof on each rhombus, its ridge a dodecahedron edge	200 − 1336√5/15

In every layer, the sum of the pieces minus the layer volume is exactly 0.

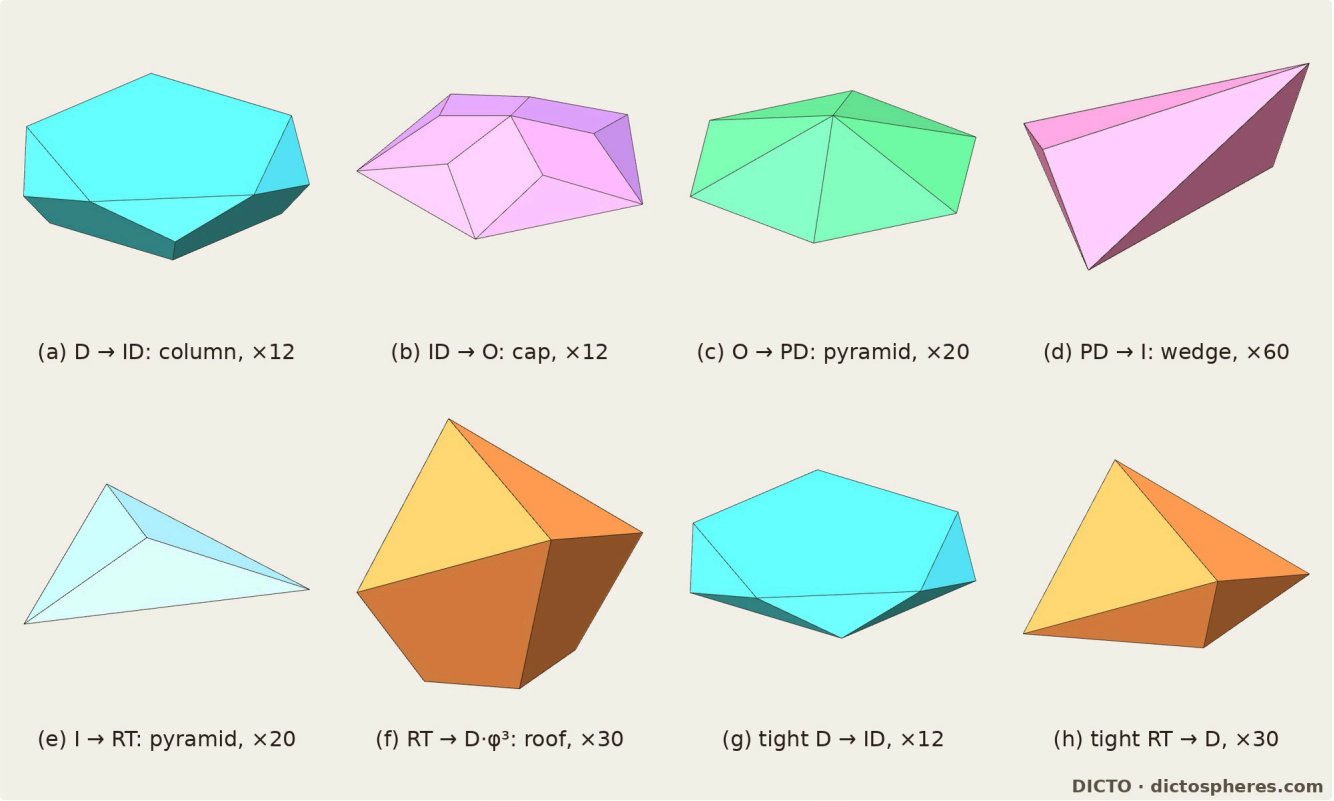


Figure 3. One piece of each layer, seen from the side with the outward direction up, each drawn at its own scale; layer colours as in the cut views. (a)–(f) the faithful nest, inside to outside (O = Ogdoecontahedron): the column on a dodecahedron face, the cap on an icosidodecahedron pentagon, the low pyramid on a regular hexagon, the irregular wedge, Kepler's pyramid, and the roof on a rhombus. (g), (h) the two end pieces that change in the tight nest (Section 4): the near-antiprism column and the wedge on a rhombus.

4. Two loose ends, and the even nest

The faithful copy keeps the original closure (outer dodecahedron = inner × φ³), and that leaves two stations floating: the triacontahedron could grow ×1.894 (float) before touching the outer dodecahedron, and the inner dodecahedron ×1.232 before touching the icosidodecahedron. The four middle stations touch. DICTO asked for a more even nest. Tightening both ends, the inner dodecahedron grows by exactly 11/18 + 5√5/18 until its corners touch the icosidodecahedron's triangles, and the outer one shrinks by exactly 5 − 2√5 until the triacontahedron's twelve tips touch its face centres. The inradii then step evenly (0.495, 0.567, 0.623, 0.653, 0.714, 0.764, 0.898), the outer layer falls from 25.2 to 1.9, and

every layer is still one kind of piece, exact with remainder 0: the inner layer becomes 12 near-antiprism columns, (−35/729 + 71√5/2187) each, and the outer 30 wedges on the rhombi, (−7352/3 + 1096√5) each. The price is the golden closure: the tight cycle repeats at linear ratio 1.8146 (squared 49815/2 − 22275√5/2), not a power of φ. We keep both versions.

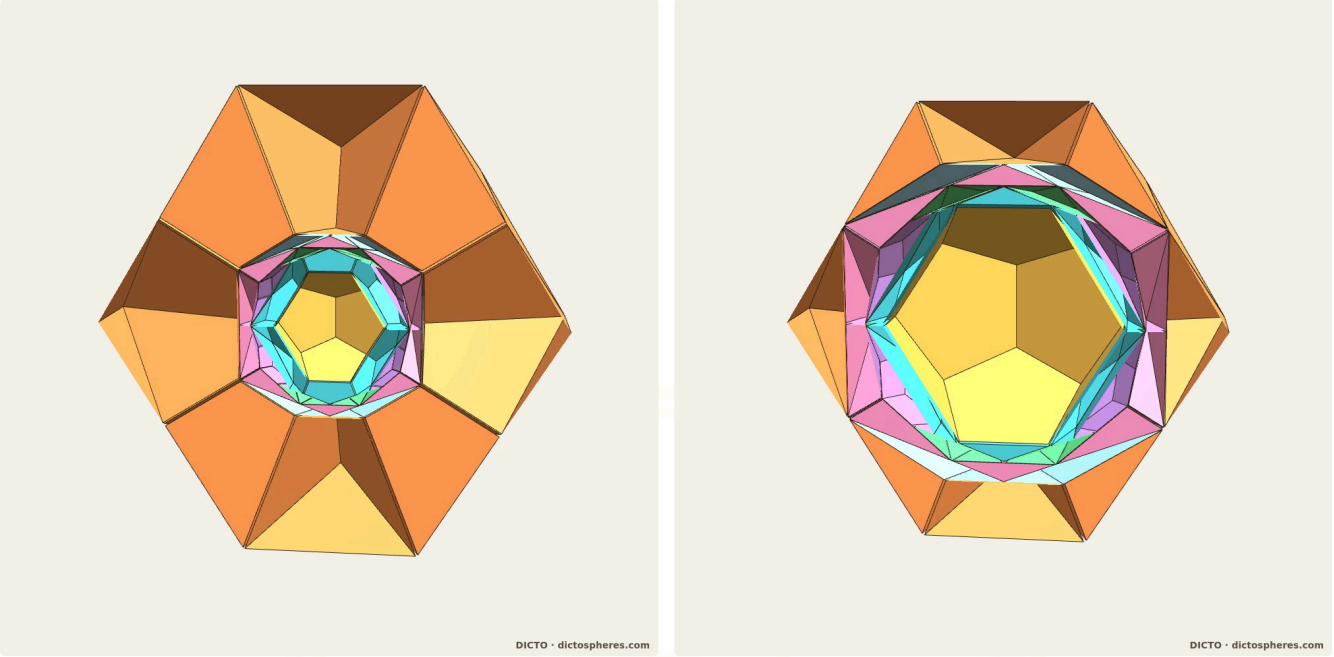


Figure 4. Faithful and tight, cut in half: (a) the faithful nest, its large outer gap; (b) the tight nest, both loose ends closed.

5. The great stellated dodecahedron in the outer gap

DICTO asked whether a Kepler–Poinsot solid could sit in the faithful nest's large outer gap. We treated each star as a filled body (points of nonzero winding number; the densities at the centre come out 3, 3, 7, 7 as they should) and found the range of sizes that keeps the triacontahedron inside the star and the star inside the outer dodecahedron. The great icosahedron does not fit. The great dodecahedron and the small stellated dodecahedron fit, but float between two sizes. The **great stellated dodecahedron fits at exactly one size**: its 20 points are the outer dodecahedron's 20 corners, and its 12 valleys, where five spikes meet, are the chain's icosahedron corners, which are also the triacontahedron's five-fold tips. We checked this exactly: each of the 12 icosahedron corners lies on five of the star's face planes.

The star is the chain's icosahedron with a spike on each face (Kepler; our In-betweens paper, DICTO DOI: 10.5281/zenodo.23345502), and its volume is exactly 3 icosahedra, as follows from the standard formulas. Its surface is 60 triangles, each seen once from the centre, and the cones over them cut the gap into two exact layers: inside the star, around the triacontahedron, **60 congruent tetrahedra** (point, triacontahedron corner and two valleys), (188/3 − 28√5) each; outside it, up to the dodecahedron, **60 congruent pentagonal pyramids**, (112/3 − 248√5/15) each. Both sum to their layers with remainder 0.

Why it fits: the squared ratio of the star's point radius to its valley radius is exactly $3\varphi^3/\sqrt{5}$, a fixed property of the great stellated dodecahedron, and the chain's outer dodecahedron and its icosahedron stand in exactly this ratio.

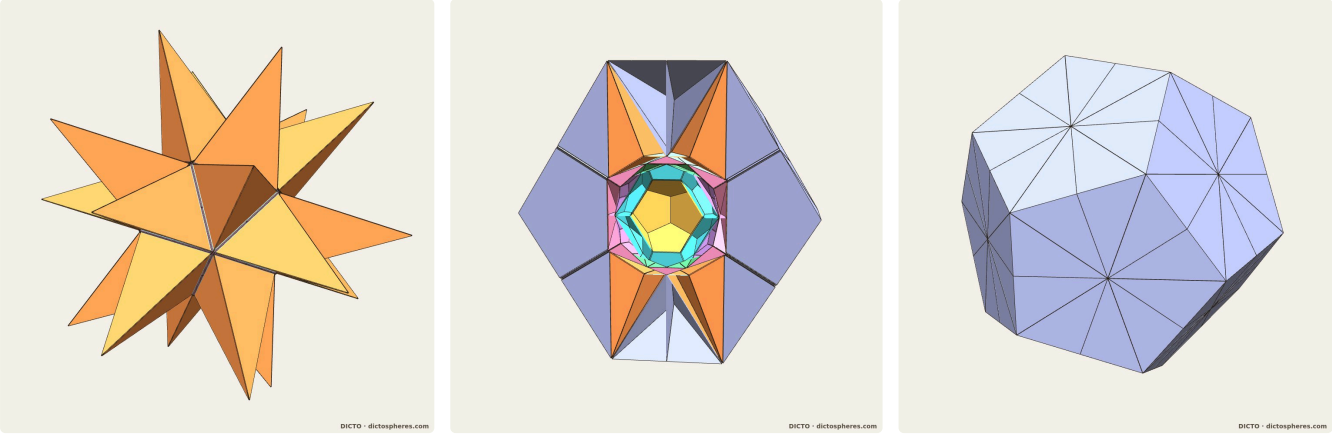


Figure 5. The great stellated dodecahedron in the outer gap: (a) the star alone, 60 tetrahedra (orange) around the triacontahedron; (b) cut in half; (c) closed by 60 pentagonal pyramids (lavender) to the outer dodecahedron.

6. The inner gap and a dual pair in reverse order

In the inner gap, between the dodecahedron and the icosidodecahedron, none of the four stars fits. The Hecatodihedron itself fits but floats: its size can vary ×1.232, touching 20 points at either end and never both. DICTO then asked for a dual pair instead. A pair X, Y locks when three contacts hold at once (dodecahedron on X, X on Y, Y on the icosidodecahedron). No icosahedral Archimedean–Catalan pair locks (the five non-chiral pairs were tested; the chiral snub dodecahedron pair was not). The truncated icosidodecahedron does *not* fit inside its dual; as DICTO pointed out, in the reversed cycle the pair must go in reverse order too, and so it does: the **disdyakis triacontahedron inside the truncated icosidodecahedron**. Exactly: the disdyakis triacontahedron (the truncated icosidodecahedron from the even permutations of $(\pm1/\varphi, \pm1/\varphi, \pm(3+\varphi))$, $(\pm2/\varphi, \pm\varphi, \pm(1+2\varphi))$, $(\pm1/\varphi, \pm\varphi^2, \pm(3\varphi-1))$, $(\pm(2\varphi-1), \pm2, \pm(2+\varphi))$, $(\pm\varphi, \pm3, \pm2\varphi)$, in the chain's golden frame, and the disdyakis triacontahedron as its polar in the unit sphere) is scaled by $9 - 3\sqrt{5} = 6/\varphi^2$, the truncated icosidodecahedron by 2/15, and the play between them is $5/9 + 2\sqrt{5}/9 = \sqrt{5}\cdot\varphi^3/9 \approx 1.0525$. The truncated icosidodecahedron's 12 decagons lie exactly in the icosidodecahedron's pentagons. The layers cut into 12 pieces of one kind, 120 of one kind and 50 caps of two kinds, all exact with remainder 0. The nearest pair to a lock is the rhombicosidodecahedron inside the deltoidal hexecontahedron, play exactly $62/123 + 137\sqrt{5}/615 \approx 1.00218$: short of locking by 0.2%.

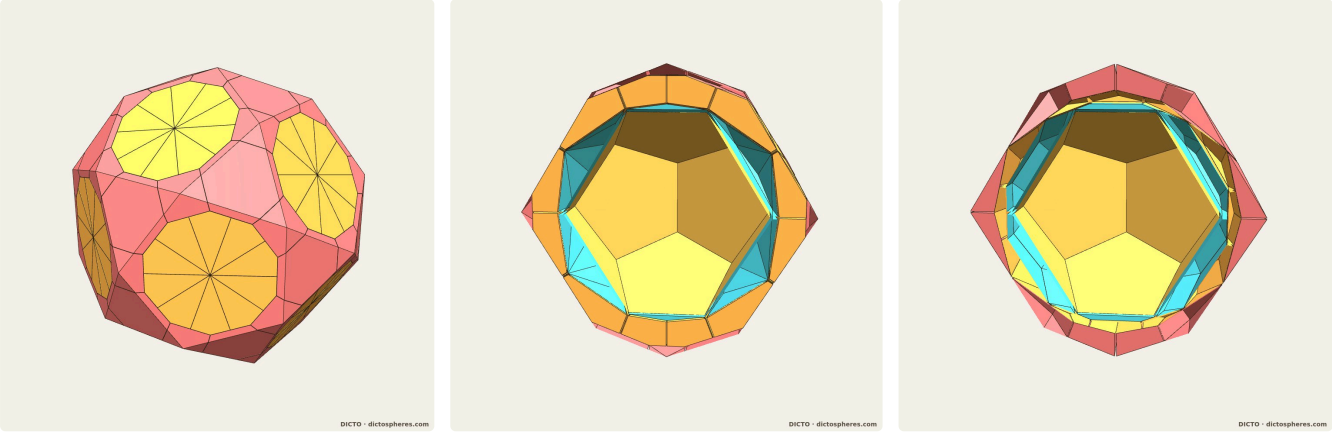


Figure 6. The inner gap: (a) the truncated icosidodecahedron pair up to the icosidodecahedron, the decagons flush in its pentagons; (b) cut in half: dodecahedron, disdyakis triacontahedron layer (cyan), truncated icosidodecahedron layer (amber), caps (coral); (c) the rhombicosidodecahedron pair, cut.

7. The golden frame

DICTO asked whether golden rectangles would frame the whole. The 15 golden rectangles of an icosahedron (five perpendicular triples) clear the whole nest when the icosahedron is the next turn of the cycle, the chain's icosahedron × φ^3 (the smallest icosahedron around the dodecahedron is not enough). Then each of the outer dodecahedron's 30 edges lies along one of the rectangles' 30 long edges, centred, and the long edge is φ^3 times as long (corner radius² $10 + 2\sqrt{5}$, midpoint radius² 4). This is classical: the 30 long edges are the edges of the outer dodecahedron's **small stellated dodecahedron**, made by extending its edges (Kepler 1619), a pentagram chord being $1 + 2\varphi = \varphi^3$ times the pentagon side. The long edges of the frame that holds the cycle are the edges of its outer dodecahedron's own first stellation.

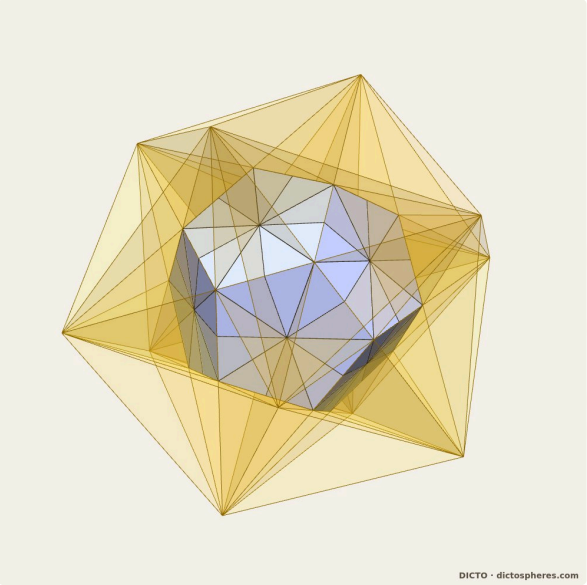


Figure 7. The nest (with the star layers) inside the 15 golden rectangles of the next turn's icosahedron.

8. The core: the cycle repeats inward

What fills the inner dodecahedron? The cycle itself. Its outer dodecahedron is exactly φ^3 times its inner one, so the whole nest scaled by $1/\varphi^3$ fills the inner dodecahedron with no gap. Inside that sits the next turn, and so on to the centre. In the faithful version every turn brings back the large gap, so the great stellated dodecahedron locks in the outer gap of every turn: its points on that turn's dodecahedron corners and its valleys holding the next triacontahedron. The golden frame of Section 7 is the next turn outward, and the core is the next turn inward.

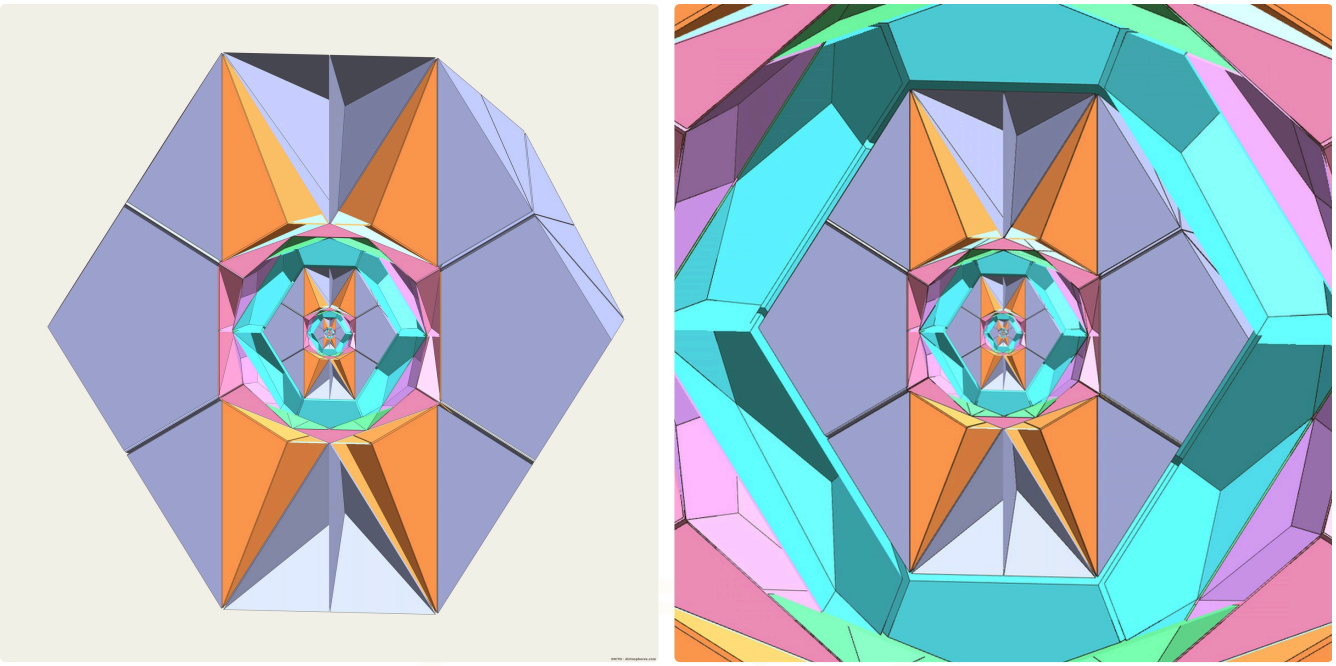


Figure 8. The infinite core, cut in half, in true proportion: (a) four turns, each a copy of the whole nest at $1/\varphi^3$ of the one around it, the great stellated dodecahedron (orange spikes) in the outer gap of every turn; (b) the centre, one turn in. Each deeper turn is its own render set in at exactly $1/\varphi^3$, so the picture is seamless and stays sharp.

The original and reversed cycles can also share one space; that is the subject of a further paper (in preparation).

9. Method: the checks this study forced on us

Three of our first tables in this study were wrong, and each was caught before publication. Standard coordinate lists for the Archimedean solids come in two mirror golden frames, and most of ours were in the wrong one, so they did not share the chain's axes. A slack formula was inverted. A float face count reported 103 kinds of piece in a layer of 120 congruent pieces. DICTO asked for rules that would prevent this, and we collected the checks in a shared module (studies/_lib/geomcheck.py) that stops with an error instead of printing a wrong table. The cone-cut script uses it directly. All the study scripts were rerun from scratch before publication, and their outputs were compared with the committed data:

1. **Frame:** every generated solid must be invariant under the chain's own five-fold and three-fold turns.
2. **Both orders:** every pair is tested both ways; polar reversal swaps the order of a dual pair.
3. **Known case first:** every formula reproduces a known answer (the dodecahedron's slack 1.232 in the icosidodecahedron; the triacontahedron locked in the star; star densities 3, 3, 7, 7) before it is used.
4. **Kinds by symmetry:** pieces are the same kind when the icosahedral group maps one onto the other, after reducing float hulls to their true corners.
5. **Exact gate:** a float result is a lead; a claim is made when the exact $\mathbb{Q}(\sqrt{5})$ remainder is 0. Exact corners come from plane triples that reproduce the point.
6. **Labels:** every number is marked exact or float, faithful or adjusted, and originals are never overwritten.

The module's tests replay each of our slips and show the check catching it.

10. Relation to known work, and status

Known and credited: polar reciprocation and the reversal of inclusion (convex geometry; Firsching 2015); the largest copy of one polytope in another (Croft 1980; Firsching 2015 computes it exactly; our fits are the concentric, same-orientation case); nested icosahedral shells in clusters, viruses and fullerenes (Takakura et al. 2007, the Tsai cluster of the Spheres of Dharma; Twarock, Valiunas and Zappa 2015; Dechant et al. 2014); the Kepler–Poinsot solids, the great stellated dodecahedron as an icosahedron with twenty spikes and the small stellated dodecahedron as the dodecahedron's extended edges (Kepler 1619; Poinsot 1810); the 15 golden rectangles of the icosahedron. Hafner (2007) dissects the small and great stellated dodecahedra into the triacontahedron and hexecontahedron using 60 tetrahedra, each, as he describes, 1/6 of a prolate golden rhombohedron; our 60 tetrahedra between the triacontahedron and the star are different pieces: each has exactly φ^2 times the volume of his on the same triacontahedron edge (checked in $\mathbb{Q}(\sqrt{5})$).

Not found in our reference library and web searches (16 October 2026; printed catalogues not read): the reversed Spheres of Dharma as an exact nest with one piece kind per layer; the tight nest; the great stellated dodecahedron's lock in the reversed cycle, which follows directly from classical facts and is presented as an observation about this cycle, not a new theorem; the inner-gap dual pairs and their exact scales and play; the exact pieces and volumes.

11. Reproducibility

In studies/2026-10-16-polar-dharma: `polar.py` (the reversed chain, exact poles, closure, touching), `polar_pieces.py` (the cone cut; kinds by symmetry orbit), `polar_exact.py` (exact corners and volumes; arguments `tight`, `tid`, `rid`), `polar_tight.py`, `polar_stars.py` and `polar_gsd_exact.py`, `polar_gsd_pieces.py`, `polar_heca_gap.py`, `polar_round.py`, `polar_dual_pair.py`, `polar_pair.py`, `polar_golden_rect.py`, `hstar_render.py`, `polar_turns.py`, `polar_hafner_exact.py`; the flip test is `flip.py` in studies/2026-10-16-flip-test; outputs in `data/`, renders by `render_polar.html`, `render_hstar.html` and the `shoot_*.mjs` scripts. Checks: `python3 studies/_lib/test_geomcheck.py`.

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